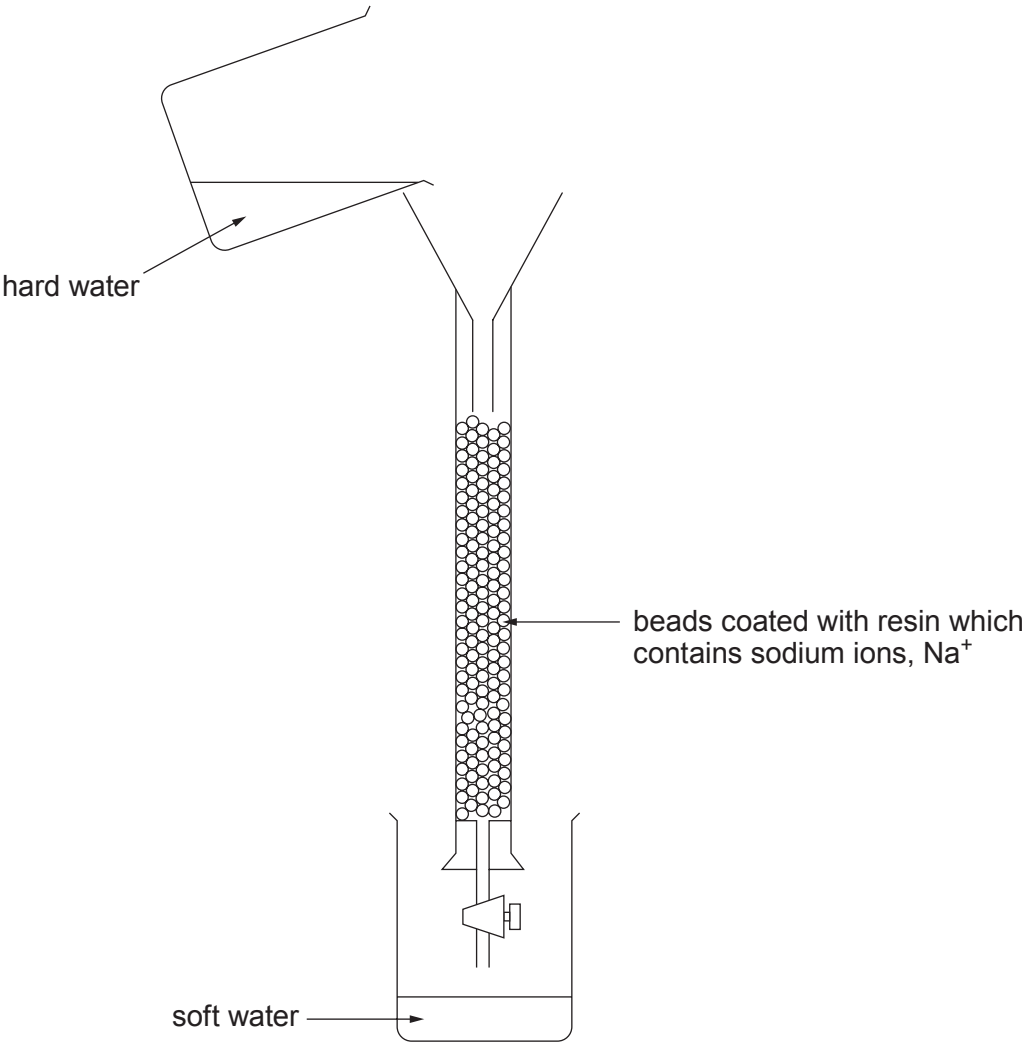


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7. (a) The diagram shows an ion exchange column used for softening permanent hard water.



(i) Explain how ion exchange works. [2]

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(ii) Explain why the ion exchange resin will stop working after continued use. Name a solution which can be passed through the resin to regenerate it. [2]

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(b) Describe and explain a method to test whether a water sample is soft water, temporary hard water or permanent hard water. [4]

(You do **not** need to include reference to fair testing in your answer)

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